MultiRate.jl: findings

Partially implicit MIS methods for the compressible Euler equations with gravity on DGSEM, terrain-following FV and cut-cell FV

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Abstract

This report collects the results of the numerical experiments with MultiRate.jl in the order they were obtained: background-free discretizations, the Agnesi hills, gravity waves, the MIS tableaus, linear acoustics with advection after the MISDG experiments, the linearized production models and a matrix-free analysis of the linearized step, Lawson RK and ETD with the MIS fast integrators, and the optimal number of substeps. The recurring result is that the slow part $S = N - L$ of a split-explicit step must not contain terms whose Courant number with the large step is large. Two instances are identified: the vertical transport of horizontal momentum by the fast vertical velocity (Agnesi a100 and the long-channel gravity wave, $U\Delta t/\Delta z = 13$–22), and a grid-scale mode on the steep flanks of the a1 hill (preliminary). The markdown version of these findings is `docs/FINDINGS.md`; the MISDG talks are in `docs/talks/`.

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1 The method

The semi-discrete system is $q' = N(q)$. At the start of a large step the fast linear operator L is frozen at q^n (in the runs it is re-frozen every 10 steps), and $S(q) = N(q) - Lq$ is the slow part. Stage i of an MIS method solves the fast system

$$Z' = LZ + \sum_{j<i} \frac{\gamma_{ij}}{d_i \Delta t}\, \Delta Y_j + \sum_{j \le i} \frac{\beta_{ij}}{d_i}\, S(Y_j), \qquad Z(0) = q^n + \sum_{j<i} \alpha_{ij} \Delta Y_j, \tag{1}$$

over $[0, d_i \Delta t]$, $d_i = \sum_j \beta_{ij}$, with $n_i = \max(1, \mathrm{round}(d_i\, n_{\mathrm{sub}}))$ substeps, and sets $Y_{i+1} = Z(d_i \Delta t)$, $\Delta Y_i = Y_{i+1} - q^n$. The paper form uses $(s+1) \times (s+1)$ matrices; MultiRate stores $\beta = P[2{:}\mathrm{end}, 1{:}\mathrm{end}-1]$ and $\alpha, \gamma = P[2{:}\mathrm{end}, 2{:}\mathrm{end}]$ (`mis_from_paper`). For $L = 0$ the method is the Runge–Kutta method with $B = (I - \alpha - \gamma)^{-1}\beta$.

The fast integrators are `:verlet` (Störmer–Verlet, thermodynamic kick and momentum), `:fb` (forward–backward), `:hevi` (horizontally explicit, vertically implicit), for FV the Strang composition of horizontal Verlet half steps around a vertical Crank–Nicolson step, `:hevi_verlet` (DGSEM), `:wrf` (WRF's off-centred vertical step with divergence damping) and, for cut cells, a Rosenbrock-W AMF step (`:rosenbrock_amf`).

The spatial discretizations are DGSEM (degree 4, LMARS/entropy-conservative volume fluxes), a terrain-following Arakawa C-grid FV with a Gal-Chen vertical coordinate, and a cut-cell C-grid FV. All production runs use MIS4 with $n_{\mathrm{sub}} = 12$.

2 Discretizations without a background state

- DGSEM uses the pointwise fast operator L (frozen $\bar\theta$ and $\gamma p/\Theta$, as in the HEVI Jacobian of Artiano et al.); terrain-following FV the conservative `:flux` form of the Θ equation; cut cells `:donor`. None of them uses $\bar\theta(z)$.
- FV without a hydrostatic reference $\bar p, \bar\rho$ (`FV_HYDRO_REFERENCE[] = false`, the default): the momentum equations use full p and ρ. The analytic hydrostatic initial state, sampled where the operator places the values (cut cells: the full-cell centre), is balanced to $O(\Delta z^2)$ (test: residual ratio > 1.9 between $n_z = 6$ and 12).
- At rest, 6 h on the a10 grid, $\max|w|$: terrain FV $4.1 \cdot 10^{-3}$ m/s (10^{-11} with the reference), cut cells $9 \cdot 10^{-3}$ m/s ($3 \cdot 10^{-5}$ with the reference).
- **Ground pressure in the Gal-Chen correction** $-\partial_\eta(z_x p)$: the ground-face p must be extrapolated from the face *means* (`fv_p_sigma`, $\frac{3}{2}p_1 - p_3 + \frac{1}{2}p_4$), so that its $O(\Delta\eta^2)$ error matches that of the interior means and cancels. Taking the lowest cell centre is off by $\rho g \Delta z/2$, an $O(1)$ error ($|U_t| \approx 0.14$ at rest, independent of resolution); pointwise 2nd/3rd-order extrapolation is only first order. The fast operator includes the ground face the same way.
- The flux form has a weak growing mode at rest on terrain (the x-flux carries θ along the sloped σ-surfaces, while the σ-correction uses 4-point averaged U; $\approx 5 \cdot 10^{-6}\,\mathrm{s}^{-1}$ on a coarse test hill, growing with horizontal resolution). It does not affect the Agnesi runs. `:advective` (state θ, adjoint pressure gradient) removes it at the price of Θ conservation.

3 Agnesi hills

Gallus & Klemp (2000): $h = 400$ m, $U = 10$ m/s, $N = 0.01\,\mathrm{s}^{-1}$, half-widths $a = 1, 10, 100$ km, periodic channel of width $40a$, $T = 21.6\,a/U$; FV 200×64 cells, DGSEM 50×16 elements of degree 4. $\max|w|$ (m/s) and the θ' range (K) at the final time:

	DGSEM `hevi_verlet`	FV terrain-following `hevi`	FV cut cells `rosenbrock_amf`
large steps	694	400	565
a1	2.77, $-1.22 \dots 0.93$	2.54, $-1.09 \dots 0.92$ ($2\Delta x$ noise)	2.14, $-1.08 \dots 0.91$
a10	0.69, $-1.48 \dots 1.65$	0.56, $-1.39 \dots 1.55$	0.58, $-1.43 \dots 1.58$
a100	0.064, $-1.49 \dots 1.70$	0.056, $-1.40 \dots 1.57$	0.057, $-1.43 \dots 1.61$

- **a100 and the vertical momentum transport.** The vertical transport of horizontal momentum by the fast vertical mass flux, $-\partial_z(\bar{u}\,W')$, must be in the implicit vertical L (`DG_VADV_FAST`, `FV_VADV_FAST`). Left in S it couples the large step to the vertical acoustics through the vertical Courant number $U\Delta t/\Delta z$ (a1 0.1, a10 1.3, a100 13 for DGSEM, 22 for FV), and a100 blows up. $\partial N/\partial q$ has no growing mode: this is a stability limit of the explicit part of MIS, not of the spatial discretization. The term is put into L for $U\Delta t/\Delta z > 4$. A split-form linearization of L (`DG_SPLIT_FAST`) makes $N - L$ vanish at rest but changes neither stability nor solution; the simple pointwise L is kept. For FV, L must in addition be skew.
- **DGSEM with `hevi_verlet`** runs all three hills at the regular horizontal acoustic step (a100: $d\tau = 26\,\mathrm{s}$, $\Delta t = 313\,\mathrm{s}$), about 16 times fewer steps than `hevi`.
- **Cut cells:** `hevi` and `rosenbrock_amf` give the same solutions; AMF also runs at the large steps of the MATLAB reference ($\Delta t = 20\,\mathrm{s}$ for a1, $100\,\mathrm{s}$ for a10). Its fast-step CFL is 0.65 of the regular-grid limit (explicit horizontal acoustics inside the W-method). Open: a warm layer of $\approx 0.35\,\mathrm{K}$ in the lowest cells downstream of the hill (a10).
- **a1, terrain FV:** $2\Delta x$ noise near the ground at the default step $\Delta t = 5.4\,\mathrm{s}$. It disappears with half the slow step (`NSUB=6`, same $d\tau$, $\max|w| = 2.41$), so it is a slow-step limit for terrain terms in $N - L$, not a spatial defect. It is also present with the hydrostatic reference and with the `:advective` form. At twice the resolution the default step blows up; `NSUB=6` is stable (2.68). See Section 8.
- Runs with $U \neq 0$ without the reference drift in mass by $\approx 2.5 \cdot 10^{-5}$ ($3.7 \cdot 10^{-7}$ with it), attributed to the sponge relaxing to a q_0 that is not a discrete equilibrium; not investigated further.

4 Gravity waves

Skamarock & Klemp (1994), flat ground:
`euler2d_gravity_wave.jl` (300 km) and `euler2d_gravity_wave_long.jl` (6000 km, $a = 100\,\mathrm{km}$, $\Delta x = 20\,\mathrm{km}$, $\Delta z = 625\,\mathrm{m}$, $U = 20\,\mathrm{m/s}$, $T = 60000\,\mathrm{s}$), MIS4, `hevi`, $n_{\mathrm{sub}} = 12$.

- 300 km: DGSEM and FV symmetric (θ' maxima $2.73/2.72 \cdot 10^{-3}\,\mathrm{K}$ DGSEM, $3.78/3.32 \cdot 10^{-3}\,\mathrm{K}$ FV).
- 6000 km, FV at the default step $\Delta t = 540\,\mathrm{s}$ ($N\Delta t \approx 5.4$, $U\Delta t/\Delta x = 0.54$, $c\Delta t/\Delta x \approx 9.5$): asymmetric packets ($\theta'$ maxima 2.2 and $9.0 \cdot 10^{-3}\,\mathrm{K}$) with grid-scale noise between them. The same with the hydrostatic reference, with the commit before the background was removed and with both fast solvers. With `NSUB=3` ($\Delta t = 135\,\mathrm{s}$, same $d\tau$) the packets are symmetric, so the error comes from the slow step. DGSEM is clean.
- **Cause and fix** (Section 7): $-\partial_z(\bar{u}W')$ was in S although $U\Delta t/\Delta z \approx 17$; the production drivers did not apply the vadv criterion of the Agnesi driver. With `set_vadv!` (now in all drivers) the default step gives θ' maxima $5.34/5.29 \cdot 10^{-3}\,\mathrm{K}$, asymmetry $4.4 \cdot 10^{-4}\,\mathrm{K}$ ($6.9 \cdot 10^{-3}\,\mathrm{K}$ before). The 300 km case ($U\Delta t/\Delta z = 2.1$) and DGSEM (0.6) are unchanged.
- The uniform $\approx 10^{-3}\,\mathrm{K}$ warm offset of θ' is also present for a 100 times smaller perturbation and for $U = 0$: a diagnostic offset of θ' against $\bar{\theta}(z)$ at the cell centres, not part of the wave.

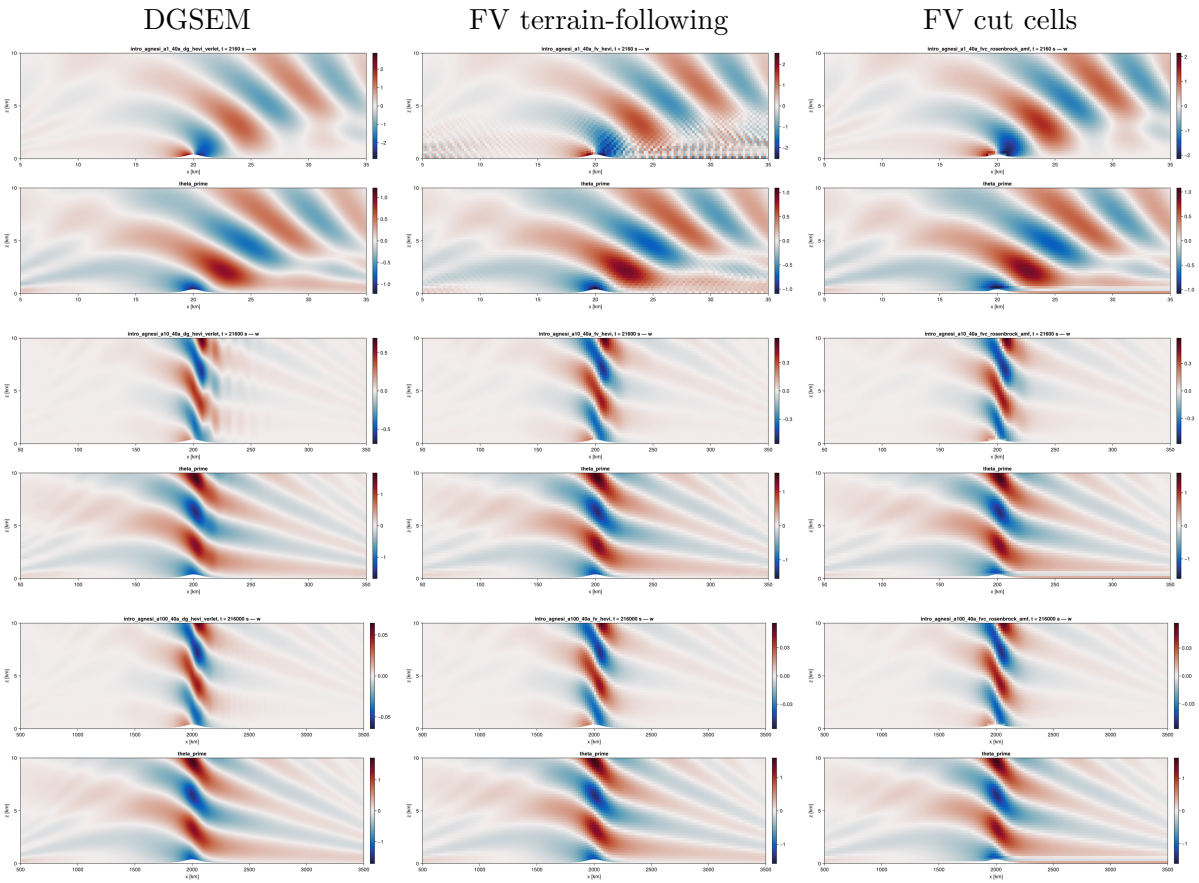


Figure 1: Agnesi hills a1 (top), a10, a100 (bottom): w and θ' at the final time.

5 MIS tableaus

`TimeStepping.mis_observed_order` computes the order with exact fast stages (augmented exponential) on a problem where L and S do not commute, $L = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$, $G(q) = (0.5 \sin q_1 + q_2^2,\ -0.3 q_1 q_2 + \cos q_1)$; `rk_order` gives the order for $L = 0$.

	WSRK3	MIS2	MIS3C	MIS4	MIS4a	MIS_KW3	MIS54
RK order ($L = 0$)	2	2	3	3	3	3	4
MIS order	2	2	**2**	3	3	3	4
fast time $\sum_i d_i$	1.83	1.19	1.19	1.44	1.47	1.00	1.88

- MIS3C (coefficients as in ClimaTimeSteppers.jl) is only second order once $L \neq 0$: a coupling condition fails. To be checked against Knoth & Wensch (2014).
- MIS_KW3 is built from the Knoth–Wolke RK3 tableau; MIS54 is from Bauer & Knoth (2021), with the coefficients of MISDG. MIS54 is fourth order and runs in the FV gravity-wave test.

6 Linear acoustics with advection

`MultiRate.LinearAnalysis` treats a linear problem $q' = Nq$ with fast part L as a model of the TimeStepping interface. `step_matrix` runs the production `mis_step!` and fast loops on unit vectors, or uses exact fast stages. A Fourier mode with complex symbol A becomes the real

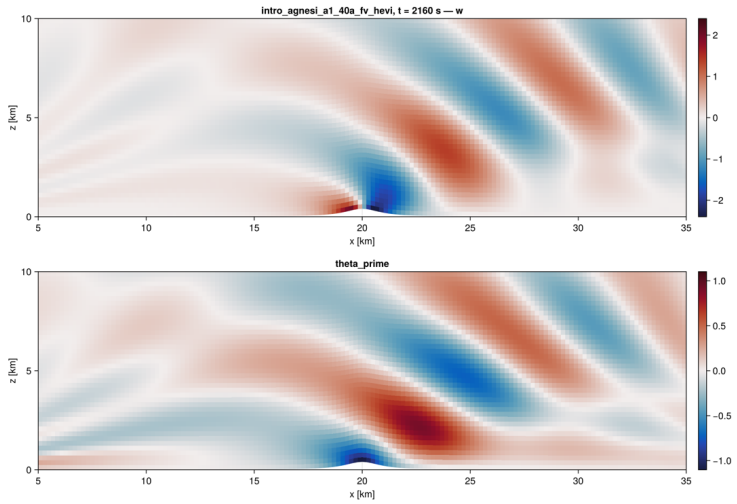


Figure 2: Terrain-following FV, a1, half the slow step (`NSUB=6`): the ground noise of the default step is gone.

model $\begin{pmatrix} \Re A & -\Im A \\ \Im A & \Re A \end{pmatrix}$. Script: `examples/linear_analysis/mis_stability.jl`. These are the experiments of the MISDG talk (`docs/talks/misdg`).

- **Knoth–Wensch test** (C-grid, 2nd-order acoustics, 3rd-order upwind advection, exact fast stages): pure advection is stable to $C_{\text{adv}} = 1.63$ (3-stage methods) and 2.06 (MIS54). Undamped WSRK3 is unstable from $C_{\text{sound}} \approx 3.2$, MIS_KW3 from ≈ 2 (and for any C_{sound} at $C_{\text{adv}} = 1$). MIS4a and MIS54 are stable on the whole box $C_{\text{sound}} \leq 12$.
- **MISDG grid** (4th-order staggered acoustics, 5th-order upwind advection, $C_{\text{adv}} = C_{\text{sound}}/7$), exact fast stages: WSRK3 stable to $C_{\text{sound}} = 2.75$, MIS4 beyond 14, MIS4a to 12, MIS54 to 14. MIS_KW3 is weakly unstable for every C_{sound} ($\rho - 1 = 7 \cdot 10^{-6}$ at $C_{\text{sound}} = 1$), as in MISDG.
- **Fast loops**, $n_{\text{sub}} = 12$: `:verlet` and `:fb` converge to the exact-stage result as $d\tau \to 0$ (orders 2 and 1). At $C_{\text{sound}} = 10$ the Verlet substep exceeds its own limit for every tableau. `:fb` makes MIS4 and MIS54 weakly unstable at $C_{\text{sound}} = 3$–7, `:verlet` does not. WRF divergence damping (`:wrf`) stabilizes WSRK3 up to $C_{\text{sound}} = 7$.
- **Pulse convergence** (1D pressure pulse, exact fast stages): orders 2 (WSRK3), 3 (MIS4a), 4 (MIS54).
- Consequence: acoustics with advection does not explain the slow-step limits of the Agnesi and gravity-wave runs; MIS4 and MIS4a are stable far beyond the Courant numbers of those runs.

MISDG grid, $C_{\text{adv}} = C_{\text{sound}}/7$	WSRK3	MIS4	MIS4a	MIS54
stable up to C_{sound} (exact fast stages)	2.75	> 14	12	14
with `:wrf`, $n_{\text{sub}} = 12$	7	7	7	7
pulse convergence order	2	3	3	4

7 Linearized production models: the long-channel failure

Script `examples/linear_analysis/fv_gravity_wave_split.jl`. The flat FV model is invariant under shifts in x. `LinearAnalysis.linearized_mode_model` builds, per horizontal Fourier mode,

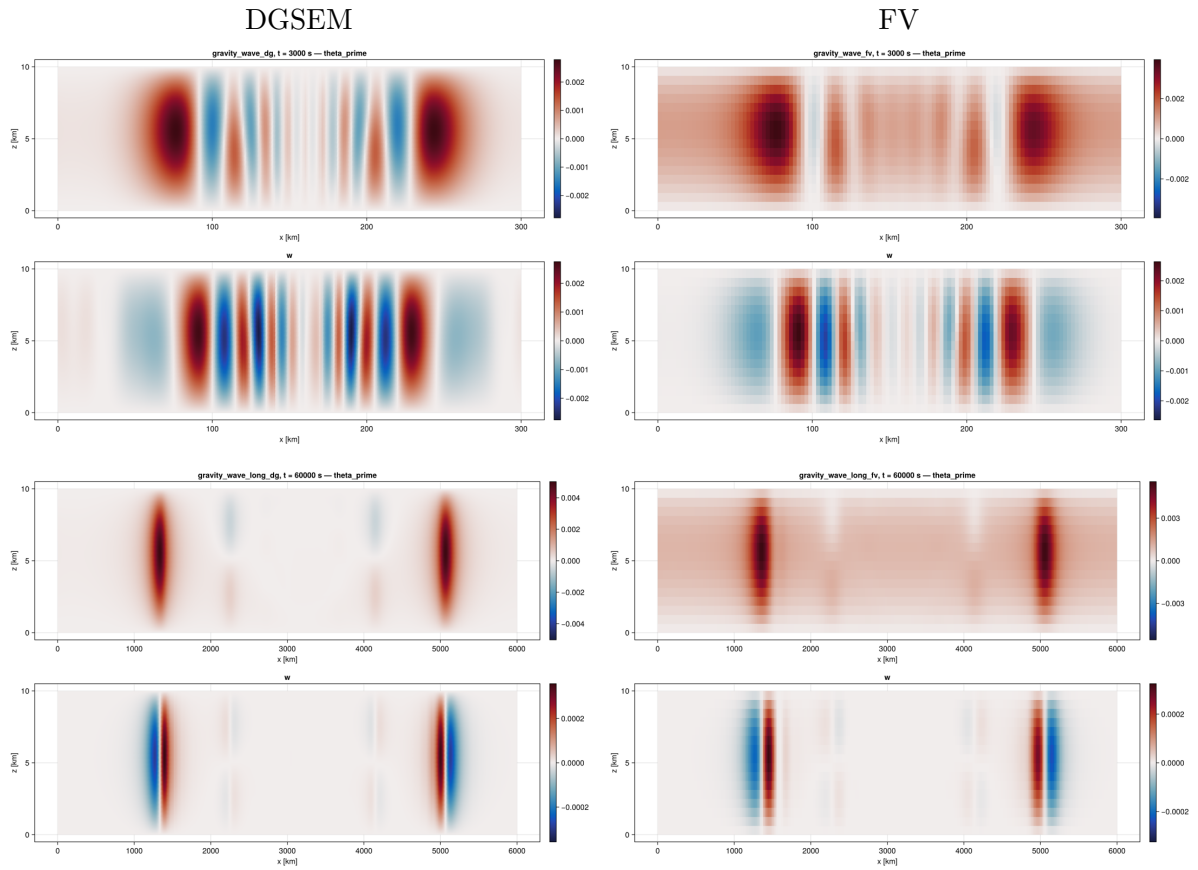


Figure 3: Gravity waves, θ' and w: 300 km channel at $t = 3000$ s (top), 6000 km channel at $t = 60000$ s (bottom; FV with the vertical momentum transport in L).

the 63×63 column blocks of the Jacobian J of `rhs_nonlinear!` (central differences), of the frozen fast operator L and of its vertical part; `step_matrix` runs the production MIS step on them. The eigenvalues of all blocks together equal those of the full fast matrix (test).

Steps that led to the cause.

1. The real FV run is linear in the amplitude: with $\Delta\theta = 10^{-4}$ instead of 10^{-2} the asymmetry scales by 10^{-2}. With $U = 0$ it vanishes (10^{-5} K).

2. A constant-coefficient model on the FV grids (`boussinesq_symbols`: compressible Boussinesq, C-grid, Lorenz grid, rigid lid, buoyancy and acoustics in L, advection in S) is accurate at $\Delta t = 540$ s (error $1.5 \cdot 10^{-4}$ of a $4.5 \cdot 10^{-3}$ K packet). Moving the advection of any rows between L and S changes this by at most 1.4 %. The failure is not in the generic acoustic–gravity–advection problem.

3. The linearized production FV reproduces the nonlinear run (θ' maxima $1.319/8.017 \cdot 10^{-3}$ K against $1.320/8.022 \cdot 10^{-3}$ K). With exact fast stages the error remains, so it is the slow-step coupling; the `:hevi` loop adds instability ($\max|\lambda| = 1.15$ at $4\Delta x$–$8\Delta x$).

4. Moving rows of S into L (exact fast stages): the horizontal-momentum rows reduce the error from $1.67 \cdot 10^{-3}$ to $2.8 \cdot 10^{-5}$ K, the Θ rows do nothing; writing L's mass and Θ fluxes with the perturbation momentum does nothing either.

5. The stiff part of the momentum rows of S is $-\partial_z(\bar{u}W')$ with coefficient $\bar{u}\Delta t/\Delta z \approx 17$, the term `FV_VADV_FAST` puts into L.

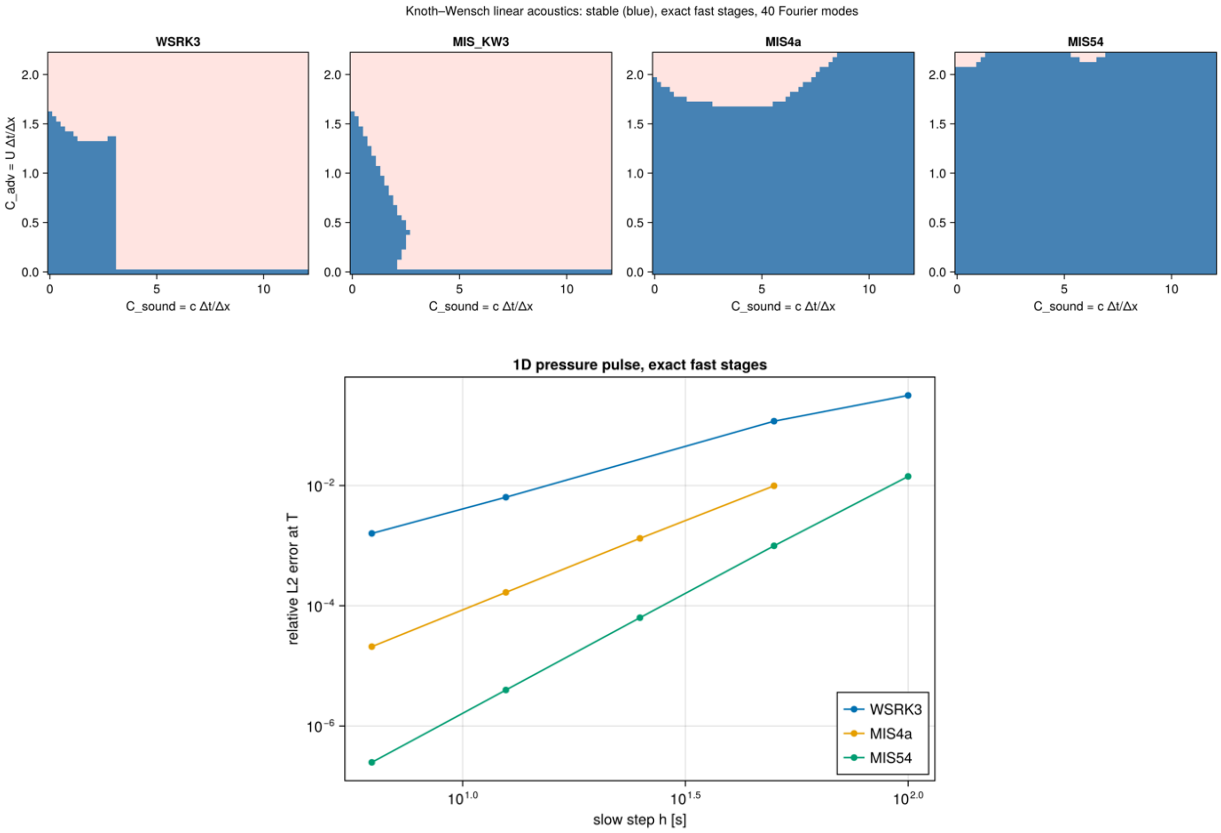


Figure 4: Top: Knoth–Wensch stability regions over $(C_{\text{sound}}, C_{\text{adv}})$, stable in blue, exact fast stages. Bottom: 1D pressure pulse, relative L^2 error against the slow step.

L	fast	Δt [s]	packets [10^{-3} K]	max err [K]	max $\lvert\lambda\rvert$
as in the driver (no vadv)	exact	540	6.13 / 2.80	$1.7 \cdot 10^{-3}$	1.0000
	`:hevi`	540	1.36 / 7.99	$4.0 \cdot 10^{-3}$	1.151
	`:hevi`	135	4.94 / 3.91	$5.6 \cdot 10^{-4}$	1.0034
vertical u-transport in L	exact	540	4.55 / 4.38	$1.2 \cdot 10^{-4}$	1.0000
	`:hevi`	540	4.42 / 4.38	$3.7 \cdot 10^{-4}$	1.0000
	`:hevi`	135	4.46 / 4.47	$2.0 \cdot 10^{-4}$	1.0000
all S momentum rows in L	`:hevi`	540	4.42 / 4.76	$4.5 \cdot 10^{-4}$	1.0015
exact linear solution			4.50 / 4.46		

MIS4, $n_{\text{sub}} = 12$ (540 s) and 3 (135 s), modes $j \le 60$ of 300; $2.8 \cdot 10^{-5}$ K is the truncation floor. Regression test: a 30-column grid with the same $\Delta x, \Delta z, \Delta t$ has max $\lvert\lambda\rvert > 1.1$ at $7.5\Delta x$ without vadv and < 1 with it.

Consequences.

- The vadv criterion ($U\Delta t/\Delta z > 4$) is a property of the step, not of a test case; every driver applies it through `set_vadv!`.
- Constant-coefficient model problems are good for the tableaus and fast loops but missed this; the linearized production operator is the reliable diagnostic.
- With `:wrf` instead of `:hevi` the default step is stable without vadv (max $\lvert\lambda\rvert = 1.0014$) but still inaccurate ($3.6/5.7 \cdot 10^{-3}$ K).

8 Matrix-free analysis of the linearized step (preliminary)

`LinearAnalysis.linearized_step(model, qb, dt, tab; nsub, fast)`
applies the linearized production MIS step, $v \mapsto (\Phi(q_b + hv) - \Phi(q_b - hv))/2h$ with the fast operator frozen at q_b, to a vector. It needs no shift invariance. `arnoldi_eigs` (restarted Arnoldi in a weighted inner product) finds eigenvalues of largest modulus of such a map.

- On the a1 terrain-FV grid (51 200 unknowns) Arnoldi on the step itself does not converge: almost all eigenvalues lie close to the unit circle. Power iteration (repeated application) works.
- a1 FV linearized about the initial state, MIS4, `:hevi`: growth **1.027 per step** at the default step (`NSUB=12`, $\Delta t = 5.4\,\mathrm{s}$) and 1.0001 (neutral, a mode spread over the domain) at `NSUB=6`.
- The growing mode is a $2\Delta x$–$3\Delta x$ oscillation of U and W in the lowest three levels on the steep flanks (slope ± 0.25) near the crest, where the noise appears in the run. A first look at $(J - L)v$ on this mode shows the largest slow tendencies at the crest in the lowest two levels in the Θ, U and W rows. A plausible estimate of a fast terrain term left in S is $c\Delta t/\Delta z \cdot |z_x| \approx 7.7 \cdot 0.25 \approx 2$.
- Open: fast loop or slow-step coupling (same Δt with many substeps), and which terrain term of S is responsible. Terrain tests are paused for now.

9 Lawson RK with the MIS fast integrators as propagator

`TimeStepping.LawsonTableau` (`:LRK4` classical RK4, `:LRK3` Kutta, `:LRK2` midpoint) is selected by name like the MIS tableaus (environment variable `TABLE` in the drivers). With the fast flow φ_τ of the frozen L (including its constant forcing) and $S = N - L$,

$$Y_i = \varphi_{c_i\Delta t}(q^n) + \Delta t \sum_j a_{ij}\, e^{(c_i - c_j)\Delta t L} S(Y_j), \qquad q^{n+1} = \varphi_{\Delta t}(q^n) + \Delta t \sum_j b_j\, e^{(1-c_j)\Delta t L} S(Y_j). \quad (2)$$

The sums are evaluated by propagating through the distinct nodes in increasing order; $\varphi_{c\Delta t}(q^n)$ is computed once per node. Since φ_τ is affine, $\varphi_\tau(x + d) = \varphi_\tau(x) + e^{\tau L} d$, so the affine fast flow can be used throughout. The propagator is the fast loop of the MIS step (`fast_solve!` without slow forcing, $d\tau \approx \Delta t/n_{\mathrm{sub}}$). `mis_step!` dispatches on the tableau type, so `run_mis`, the drivers and `LinearAnalysis` work unchanged. Fast integration time per step: LRK4 $2.5\Delta t$, MIS4 $1.44\Delta t$.

- **Order** with exact propagators (non-commuting test): LRK2 2, LRK3 3, LRK4 4. With the Verlet loop the step converges to the exact one at order 2 in $d\tau$.
- **Linear stability** is limited by the advection Courant number, as for RK4: $C_{\mathrm{adv}} \le 1.75$ (LRK4), 1.62 (LRK3) on the Knoth–Wensch test (MIS54: 2.06); stable fraction of the box 0.78 (MIS54: 0.99). On the MISDG grid ($C_{\mathrm{adv}} = C_{\mathrm{sound}}/7$) LRK4 with Verlet and $n_{\mathrm{sub}} = 12$ is stable up to $C_{\mathrm{sound}} = 10$, where MIS4 and MIS54 with Verlet are already unstable; $|\rho| - 1 \approx 10^{-8}$ against 10^{-5} for MIS4.
- **Accuracy with exact propagators:** the pulse error is 190 times smaller than with MIS54 at the same step ($5.3 \cdot 10^{-6}$ against $9.9 \cdot 10^{-4}$), order 4. Lawson is exact for $S = 0$; MIS with $\alpha, \gamma \neq 0$ is not. For an unbalanced initial state (the gravity-wave θ' at constant pressure) all MIS tableaus damp or dephase the fast acoustic part by a Δt-independent amount ($4 \cdot 10^{-3}$ relative in the linear Boussinesq wave, also for the $k = 0$ mode); Lawson keeps it ($5 \cdot 10^{-13}$).
- **Accuracy with the real fast loops:** the propagator error enters directly. Pulse with Verlet: the LRK4 error is the second-order Verlet error (rates 2.00), the same as for MIS54. Nonlinear FV gravity wave (300 km, `:hevi`) against explicit RK4 on N:

$\max \lvert\theta' - \theta'_{\mathrm{RK4}}\rvert$ at $z = 5\,\mathrm{km}$ [K]	$\Delta t = 67\,\mathrm{s}$	34 s	17 s
MIS4	$1.1 \cdot 10^{-3}$	$8.6 \cdot 10^{-4}$	$6.0 \cdot 10^{-4}$
MIS54	$9.4 \cdot 10^{-4}$	$7.9 \cdot 10^{-4}$	$5.7 \cdot 10^{-4}$
LRK4	$1.6 \cdot 10^{-3}$	$1.7 \cdot 10^{-3}$	$3.7 \cdot 10^{-4}$

($\max \theta'_{\mathrm{RK4}} = 4.4 \cdot 10^{-3}\,\mathrm{K}$, $n_{\mathrm{sub}} = 12$.) All methods are dominated by the fast-loop error (vertical Crank–Nicolson at large vertical Courant numbers).

Consequence (revised in Section 10). With second-order fast loops the Lawson error at small slow steps is the fast-loop error. The table mixes the gravity wave with an oscillation of the horizontal mean; at large slow steps and with a fourth-order fast loop the picture is different. A MIS run with a small step is not a neutral reference for Lawson (MIS's own acoustic damping); RK4 on N is.

10 ETD, high-order fast loops, and MIS against Lawson/ETD at large slow steps

ETD in stage-ODE form (`ETDTableau`, after the MATLAB `SplitExplicitETD.m` and `MethodETD.m`). Stage i starts from q^n and integrates, with $T = c_i \Delta t$ and $\tau = t/T$,

$$Z' = LZ + \sum_{j \le i} \sum_{k} P_{ijk}\, \tau^{k-1} S(Y_j), \qquad P_{ijk} = \frac{a_{ijk}}{(k-1)!\, c_i}, \tag{3}$$

which, exactly integrated, is the ETD stage $e^{TL} q^n + \Delta t \sum_{j,k} a_{ijk} \varphi_k(TL) S(Y_j)$. Tableaus: EB1 (exponential Euler), ETD2 (Cox–Matthews), EB4 (Krogstad's ETDRK4, $c = \frac{1}{2}, \frac{1}{2}, 1, 1$, fast time $3\Delta t$). `fast_solve_poly!` runs the fast loops with the forcing at the midpoint of every base step, which keeps symmetric loops symmetric in (q, t). Orders with exact fast stages: 1, 2, 4.

High-order fast loops. Symmetric compositions of a symmetric second-order loop (`:verlet`, FV `:hevi`, `:hevi_verlet`): `_y4` (Yoshida triple jump, weights $1.35, -1.70, 1.35$), `_s4` (Suzuki, 5 stages, $|w| \le 0.66$, stable at the regular substep), `_y6` (Yoshida, 7 stages). Measured orders 4, 4, 6, also with the time-dependent ETD forcing; Suzuki's error constant is 60 times smaller than the triple jump's.

Linear pulse at large steps ($h = 50\,\mathrm{s}$, $C_{\mathrm{sound}} = 7$, $C_{\mathrm{adv}} = 1$, relative error at T):

	exact fast stages	Verlet, $n_{\mathrm{sub}} = 48$	Suzuki, $n_{\mathrm{sub}} = 8$
MIS4a	$9.9 \cdot 10^{-3}$	$9.9 \cdot 10^{-3}$	$9.9 \cdot 10^{-3}$
MIS54	$9.9 \cdot 10^{-4}$	$8.4 \cdot 10^{-4}$	$1.0 \cdot 10^{-3}$
EB4	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-3}$
LRK4	$\mathbf{5.3 \cdot 10^{-6}}$	$3.1 \cdot 10^{-4}$	$\mathbf{5.8 \cdot 10^{-6}}$

Nonlinear FV gravity wave at large steps (300 km, reference: explicit RK4 on N). The initial θ' at constant pressure is not balanced and excites a vertical acoustic oscillation of the horizontal mean of θ' ($\approx 10^{-3}\,\mathrm{K}$ at $z = 5\,\mathrm{km}$). RK4, Lawson and ETD keep it, MIS damps it. The wave error (horizontal mean removed) at the best number of substeps:

wave error [K]	$\Delta t = 67\,\mathrm{s}$ ($U\Delta t/\Delta x = 0.53$)	130 s (1.04)	200 s (1.6)
MIS4 `hevi`	$2.2 \cdot 10^{-4}$	$5.6 \cdot 10^{-4}$	unstable
MIS54 `hevi`	$7.5 \cdot 10^{-5}$	$\mathbf{1.0 \cdot 10^{-4}}$	$\mathbf{3.2 \cdot 10^{-4}}$
LRK4 `hevi_s4`	$7.2 \cdot 10^{-5}$	$2.0 \cdot 10^{-4}$	unstable
EB4 `hevi_s4`	$\mathbf{6.3 \cdot 10^{-5}}$	$1.8 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$

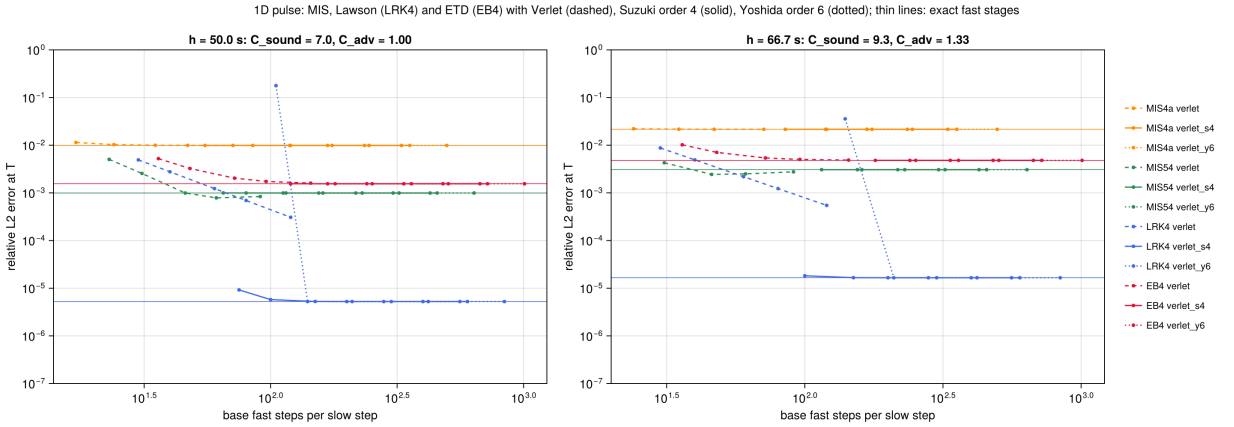


Figure 5: Linear pulse: error against base fast steps per slow step.

With the mean, MIS has 0.9–$1.4 \cdot 10^{-3}$ K at all steps; LRK4 and EB4 with `hevi_s4` 6–$9 \cdot 10^{-5}$ (67 s) and 1.6–$2.1 \cdot 10^{-4}$ (130 s); with the second-order `hevi` their total error is erratic (10^{-4} to $1.7 \cdot 10^{-3}$).

MIS against Lawson/ETD.

- MIS solves each stage as $Z' = LZ +$ constant forcing; it is an ETD-type method whose φ-functions come from the stage structure. With $\alpha, \gamma \neq 0$ it is not exact for $S = 0$ and damps or dephases fast modes at $\omega \Delta t \gg 1$ by a Δt-independent amount. Lawson and ETD are exact for $S = 0$ and keep the fast modes.
- Where the fast modes are the signal (the acoustic pulse), Lawson with an accurate propagator is far better (170 times below MIS54). ETD (EB4) is not; a plausible reason is that its polynomial-in-time forcing interpolates S along the fast oscillation, while Lawson interpolates in the frame moving with e^{tL}.
- Where the slow modes are the signal (the gravity wave), MIS54 is as accurate as LRK4 and EB4 at $U \Delta t / \Delta x \approx 0.5$, more accurate at ≈ 1, and the only one of them stable and accurate at 1.6. It needs no high-order fast loop, so it is 5–8 times cheaper per step.

Optimal number of substeps.

- MIS: once the substep is stable the error depends neither on the fast loop (`hevi`, `hevi_s4`, `hevi_y6` give the same) nor on n_{sub}. The optimum is the smallest stable n_{sub}, $d\tau$ at the acoustic limit, with at most a margin of about 1.5 (MIS54 at 67 s: $1.1 \cdot 10^{-4}$ at $n_{\text{sub}} = 12$, $7.1 \cdot 10^{-5}$ at 24). A higher-order fast loop is wasted.
- Lawson/ETD: the wave error behaves like MIS, but the fast part, and with it the total error, needs a fourth-order loop. `_s4` at the smallest stable n_{sub} reaches the exact-propagator error; more substeps or `_y6` bring nothing. Cost: 5 times the second-order loop. With a second-order loop the error falls only like $(\Delta t / n_{\text{sub}})^2$.

11 EPIRK(-W) and exponential Adams–Bashforth with the MIS fast loops

Methods. Both are written in the stage-ODE form of Section 10: the φ-function combinations that KIOPS evaluates in the EPI/EPIRK codes of Tokman and Gaudreault become fast integrations with a polynomial forcing.

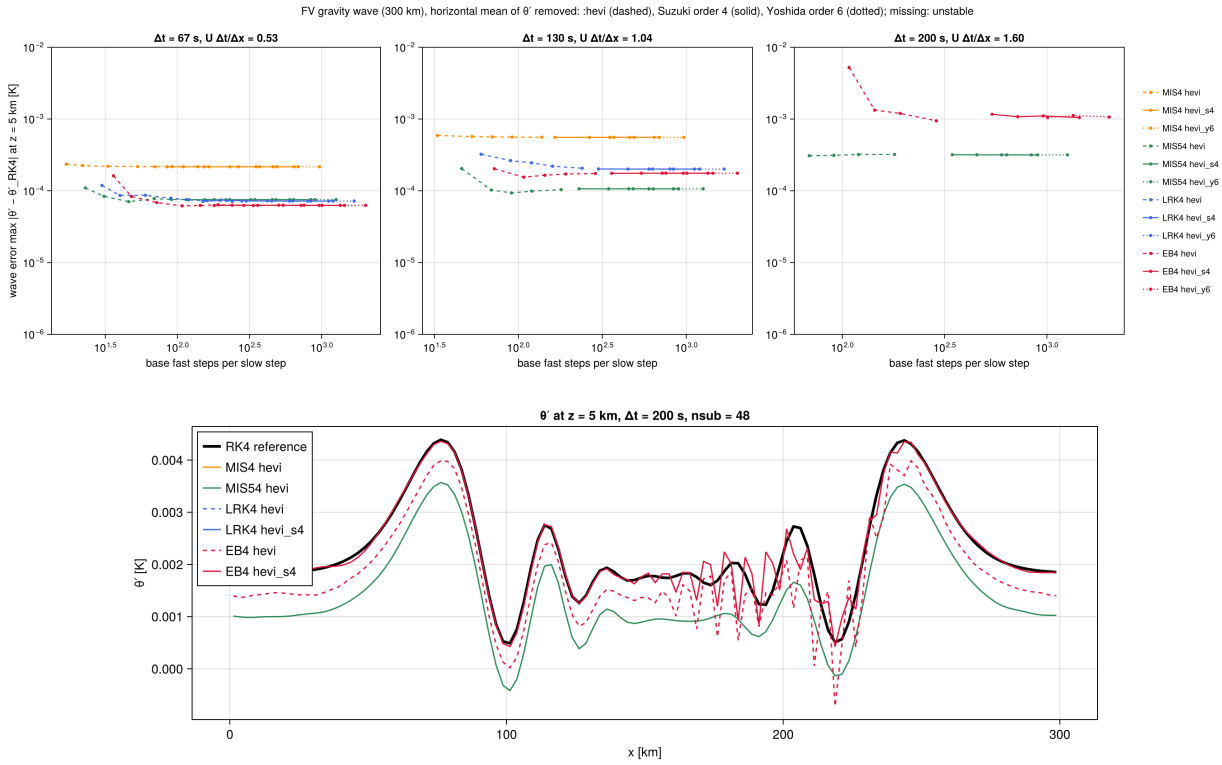


Figure 6: FV gravity wave. Top: wave error against base fast steps per slow step. Bottom: θ' at $z = 5\,\mathrm{km}$ for $\Delta t = 200\,\mathrm{s}$; the MIS54 curve is shifted by the damped mean oscillation, EB4 with `hevi` shows grid-scale noise.

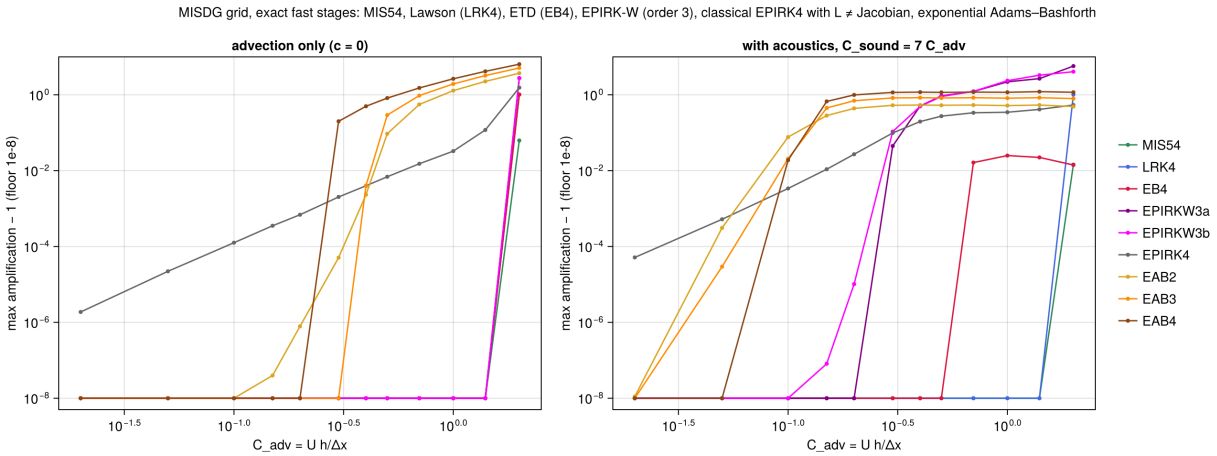
- EPIRK (`EPIRKTableau`): $Y_i = q^n + a_{i1}\psi_1(g_{i1}hL)\,hN(q^n) + \sum_{j\geq 2} a_{ij}\psi_j(g_{ij}hL)\,h\,\Delta^{(j-1)}r$, $\psi_j = \sum_k p_{jk}\varphi_k$, $r(y) = N(y) - N(q^n) - L(y - q^n) = S(y) - S(q^n)$. For each distinct $g > 0$ of a row one fast integration from q^n over gh of $W' = LW - Lq^n + \sum_k \tau^{k-1}G_k$, $G_k = \sum_j a_{ij}p_{jk}V_j/((k-1)!\,g)$; terms with $g = 0$ are explicit. Tableaus of Narayanamurthi, Tranquilli, Sandu, Tokman (2019): `EPIRKW3a`, `EPIRKW3b` (EPIRK-W, order 3 for any L; fast time $2.27h$ and $1.35h$) and `EPIRK4` (classical order 4, only for $L = \partial N/\partial q$).
- Exponential Adams–Bashforth (`EABTableau`, `EAB2`–`EAB4`), the multistep form of Gaudreault's EPI methods for an arbitrary L: one fast integration per step, $Y' = LY + p(\tau)$ from q^n over h, p the polynomial through $S(q^{n-m})$ at $\tau = -m$, all S with the current frozen L (the history stores q and N). Start-up with EB4.

Orders (exact fast stages, $L \neq$ Jacobian): EPIRKW3a/b 3.0, EAB2/3/4 2.0/3.0/3.95; EPIRK4 drops to 0.97 and has 3.9 with $L =$ Jacobian. The production step agrees with the exact φ-functions and converges with order 4 with the Suzuki loop.

Stability (MISDG grid, exact fast stages, $C_{\mathrm{sound}} = 7C_{\mathrm{adv}}$, maximum amplification):

C_{adv}	0.1	0.2	0.3	0.5	1.0
MIS54, LRK4	1	1	1	1	1
EB4	1	1	1	1	1.025
EPIRKW3a / W3b	1	1	1.04 / 1.11	1.9	3.2 / 3.4
EPIRK4 ($L \neq J$)	1.003	1.03	1.10	1.27	1.35
EAB2 / 3 / 4	1.08 / 1.02 / 1.02	1.4 / 1.7 / 2.0	1.5 / 1.8 / 2.2	1.5 / 1.8 / 2.2	1.5 / 1.8 / 2.2

Without acoustics EPIRK-W is stable up to $C_{\mathrm{adv}} = 1.4$ and EAB2/3/4 up to the Adams–Bashforth limits $\approx 0.4/0.4/0.2$: the instability comes from the coupling with the fast acoustics.



Nonlinear FV gravity wave (300 km, wave error in K, `hevi`, n_{sub} at the acoustic limit):

Δt [s]	67	33	20	10	5	2.5	1
$c\Delta t/\Delta x$	9.3	4.7	2.8	1.4	0.7	0.35	0.14
MIS54	1.1e-4	1.1e-4	1.1e-4	7.0e-5			4.2e-6
EB4	1.6e-4	1.2e-4	9.1e-5	9.0e-5	8.8e-5	3.6e-5	4.2e-6
EPIRKW3a	unst.	unst.	1.2e-2	unst.			
EPIRKW3b	unst.	unst.	unst.	8.4e-5	unst.	3.6e-5	4.2e-6
EPIRK4	8.0e-4	2.7e-3	4.3e-4	1.6e-4			
EAB3	unst.	unst.	unst.	unst.	unst.	3.7e-5	4.2e-6

All converge to the same solution at small steps, but EPIRK-W and EAB are usable only where the acoustics are almost resolved ($c\Delta t/\Delta x \lesssim 0.5$), which defeats the multirate idea.

Why.

- The W-order of EPIRK-W is a classical (non-stiff) order for an arbitrary matrix; nothing controls the method for $\|hL\| \gg 1$ (Narayanamurthi et al. note that $A = J$ may be needed for stability). In their applications A approximates the full Jacobian.
- EAB extrapolates S from previous steps. Through the coupling with advection S carries the fast oscillation, and polynomial extrapolation of $e^{i\omega t}$ with $\omega h \gg 1$ amplifies it. EB4 interpolates only inside the step, Lawson in the frame moving with e^{tL}.
- The classical EPIRK4 with L instead of the Jacobian is first order; the real Jacobian would need a Jacobian-vector product of the slow part in every fast substep.

Consequence. Keep Lawson (LRK4) and ETD-RK (EB4) next to MIS. A multistep method with one slow evaluation per step would need the Lawson form (integrating-factor Adams–Bashforth: the stored S propagated by the fast flow; k fast integrations per step), not tried yet.

12 MIS with a frozen part of the right-hand side (flat FV)

Idea (Knoth & Wensch, *B-series and split-explicit methods*; MATLAB `SplitExplicit.m`). Instead of adding and subtracting a linearized fast operator, the right-hand side is written with two arguments, $N(q) = F(q, q)$, the first frozen at the stage values, the second evolving in the fast integration,

$$Z_i' = \sum_j w_{ij} F(Y_j, Z_i) + \sum_j \frac{\gamma_{ij}}{d_i \Delta t}(Y_{j+1} - q^n), \quad w_{ij} = \beta_{ij}/d_i,$$

over $d_i\Delta t$ from $q^n + \sum_j \alpha_{ij}(Y_{j+1} - q^n)$ (`run_frozen`). For an additive $F(y,z) = S(y) + G(z)$ this is the usual MIS step.

FV split (flat grids). Fast, evolving z: $\rho_t = -\partial_x U - \partial_z W$, $\Theta_t = -\partial_x(\theta_x U) - \partial_z(\theta_z W)$, $U_t = -\partial_x p'(\Theta)$, $W_t = -\partial_z p'(\Theta) - g\rho'$ with the full nonlinear $p(\Theta)$, and the sponge. Frozen coefficients $C(y)$: the upwinded face values θ_x, θ_z of the full operator (linear in C, so $\sum_j w_j F(Y_j, Z) = F_f(\sum_j w_j C(Y_j), Z) + \sum_j w_j S(Y_j)$). Slow, frozen y: momentum advection. $F(q,q) = N(q)$ to rounding; nothing is linearized and no reference state is needed. Fast loops: `verlet`, `fb`, `rk4`, compositions, and `hevi` (horizontal Verlet half steps around a linearly implicit vertical Crank–Nicolson step $z \leftarrow z + (I - \frac{dt}{2} J_v)^{-1} dt\, G_v(z)$, G_v with the nonlinear $p(\Theta)$, J_v with $a = \gamma p/\Theta$ from the start of the stage).

What is frozen. With the two-argument form $F(y,z)$, y the stage values and z the state of the fast integration, the flat FV split freezes exactly two things:

1. the face values of θ in the Θ fluxes: the third-order upwind values θ_x, θ_z of the full operator, computed from $\theta = \Theta/\rho$ of y with the upwind direction from U, W of y; within stage i the weighted sum $\sum_j (\beta_{ij}/d_i)\, \theta_{\text{face}}(Y_j)$;
2. the momentum advection $\partial_x(Uu) + \partial_z(Wu)$, $\partial_x(Uw) + \partial_z(Ww)$, evaluated once per stage at Y_j and added as a constant forcing (with the same weights and the γ terms of MIS).

Everything else is evaluated at the current z in every substep: the mass fluxes U, W, the Θ fluxes $\theta_{\text{face}}U$, $\theta_{\text{face}}W$ with the current U, W, the pressure $p(\Theta)$ from the nonlinear equation of state, gravity $g\rho$ and the sponge. For $y = z = q$ this is $N(q)$.

Remarks.

- Compared with the linearized split nothing of the acoustics is frozen: no $a = \gamma p/\Theta$, no averaged $\bar{\theta}$, no difference $S = N - L$. The frozen data are values of the full operator, so the split is exact and needs no reference state.
- The only linearization left is the Jacobian of the implicit vertical Crank–Nicolson step of `hevi`: $a = \gamma p/\Theta$ at the start of the stage, used only in $(I - \frac{dt}{2} J_v)^{-1}$; the vertical tendency itself uses the nonlinear $p(\Theta)$. With `verlet` nothing is linearized.
- Freezing θ_{face} makes F non-additive: $F(y,z) \neq S(y) + G(z)$, because the Θ flux is a product of a frozen (θ) and a live (U, W) factor. This produces the mixed elementary differentials ($F^{(2)}_{(1,1)}FF$, ...) of the B-series and is the reason why MIS4, MIS4a, MIS54 drop to order 2 in this form.
- The freezing is linear in the frozen data (Θ flux linear in θ_{face}, momentum advection additive), so $\sum_j w_j F(Y_j, Z) = F_f(\sum_j w_j C(Y_j), Z) + \sum_j w_j S(Y_j)$: one fast operator per stage, as cheap as one evaluation of the fast part. Freezing a nonlinear function of y (e.g. the upwind direction inside the fast loop) would need F_f for every Y_j.
- Alternative: freeze only the momentum advection and compute θ_{face} from z. The split is then additive, $F = S(y) + G(z)$, and the MIS orders are kept (MIS54: 4), but the fast part contains the nonlinear upwind transport of θ (costlier per substep, and its vertical part is harder to treat implicitly).
- Other choices are possible in the same framework, e.g. freezing the velocity of the momentum transport (frozen u, live U: transport of the fast momentum by the slow wind), which would move part of the advection into the fast loop like the vertical transport term of `FV_VADV_FAST`.

Frozen pressure. Two further options freeze the pressure as well (`FV_FROZEN_PRESSURE`); the fast part is then linear in z:

- `:derivative`: $\nabla p = (\mathrm{d}p/\mathrm{d}\Theta)(y)\,\nabla\Theta_z$ with $\mathrm{d}p/\mathrm{d}\Theta$ frozen at the faces. It is taken as the discrete derivative $(p_r - p_l)/(\Theta_r - \Theta_l)$ of the two neighbouring cells ($\gamma p/\Theta$ at the mean Θ if they are nearly equal), so that it reproduces the pressure difference of N and $F(q,q) = N(q)$ holds exactly.
- `:ratio`: $p = (p/\Theta)(y)\,\Theta_z$ with p/Θ frozen at the centres ($F(q,q) = N(q)$ trivially).

With a linear vertical part the implicit step of `hevi` is exactly Crank–Nicolson; the column solver is written for a pressure difference $\alpha_k\Theta_k - \beta_k\Theta_{k-1}$ at the z-face k ($\alpha_k = \beta_k = (\mathrm{d}p/\mathrm{d}\Theta)_k$, resp. $\alpha_k = (p/\Theta)_k$, $\beta_k = (p/\Theta)_{k-1}$).

Results (gravity wave, wave error in K, `hevi` at the substep limit):

Δt [s]	67	100	130	167	200
MIS54, $p(\Theta_z)$	1.1e-4	9.6e-5	1.6e-4	1.2e-4	1.6e-4
MIS54, $\mathrm{d}p/\mathrm{d}\Theta$ frozen	1.2e-4	1.0e-4	2.1e-4	1.9e-4	2.6e-4
MIS54, p/Θ frozen	unst.	unst.	unst.	unst.	unst.
MIS54, linearized L	1.1e-4	9.6e-5	2.0e-4	2.4e-4	3.1e-4
MIS4, $\mathrm{d}p/\mathrm{d}\Theta$ frozen	2.6e-4	3.3e-4	3.8e-4	4.2e-4	1.6e-3

The `:ratio` form converges (MIS54: $3.3\cdot10^{-7}$, $1.3\cdot10^{-6}$, $4.6\cdot10^{-6}$ at $\Delta t = 0.5, 1, 2\,\mathrm{s}$) but is stable only for $c\Delta t/\Delta z \lesssim 1$ (unstable from $\Delta t = 5\,\mathrm{s}$, $\Delta z = 625\,\mathrm{m}$).

Why. What is frozen must be non-stiff as a function of y: the derivative of F with respect to its first argument, $F_{(1,0)}$, is treated explicitly over the whole slow step. With $p = (p/\Theta)(y)\,\Theta_z$ a perturbation $\delta\Theta$ changes the fast pressure only by $(p/\Theta)\,\delta\Theta = \gamma^{-1}(\mathrm{d}p/\mathrm{d}\Theta)\,\delta\Theta$; the rest, $(\gamma - 1)(p/\Theta)\,\delta\Theta$, enters through the frozen coefficient, $F_{(1,0)} \ni \Theta\,\partial_\Theta(p/\Theta)\,\delta\Theta$, i.e. about 30 % of the acoustic stiffness is explicit and the fast loop carries sound waves with the wrong speed $c/\sqrt{\gamma}$. With $\mathrm{d}p/\mathrm{d}\Theta$ frozen the fast loop has the full acoustic stiffness and the frozen coefficient varies only with $\mathrm{d}^2p/\mathrm{d}\Theta^2\,\nabla\Theta$, a smooth slow quantity. The same criterion explains why freezing θ_{face} is harmless: θ is advected, its variation is slow. Rule: *freeze coefficients, not parts of the stiff response.* The frozen $\mathrm{d}p/\mathrm{d}\Theta$ costs a little accuracy against the nonlinear $p(\Theta_z)$ (up to 1.6 times at 200 s, still better than the linearized split) and makes the fast part linear, which is what an implicit vertical solver or a DG/HDG fast operator needs.

Orders (toy oscillator with frozen coefficient $c(y) = 1 + 0.5\sin y_1$, exact fast stages):

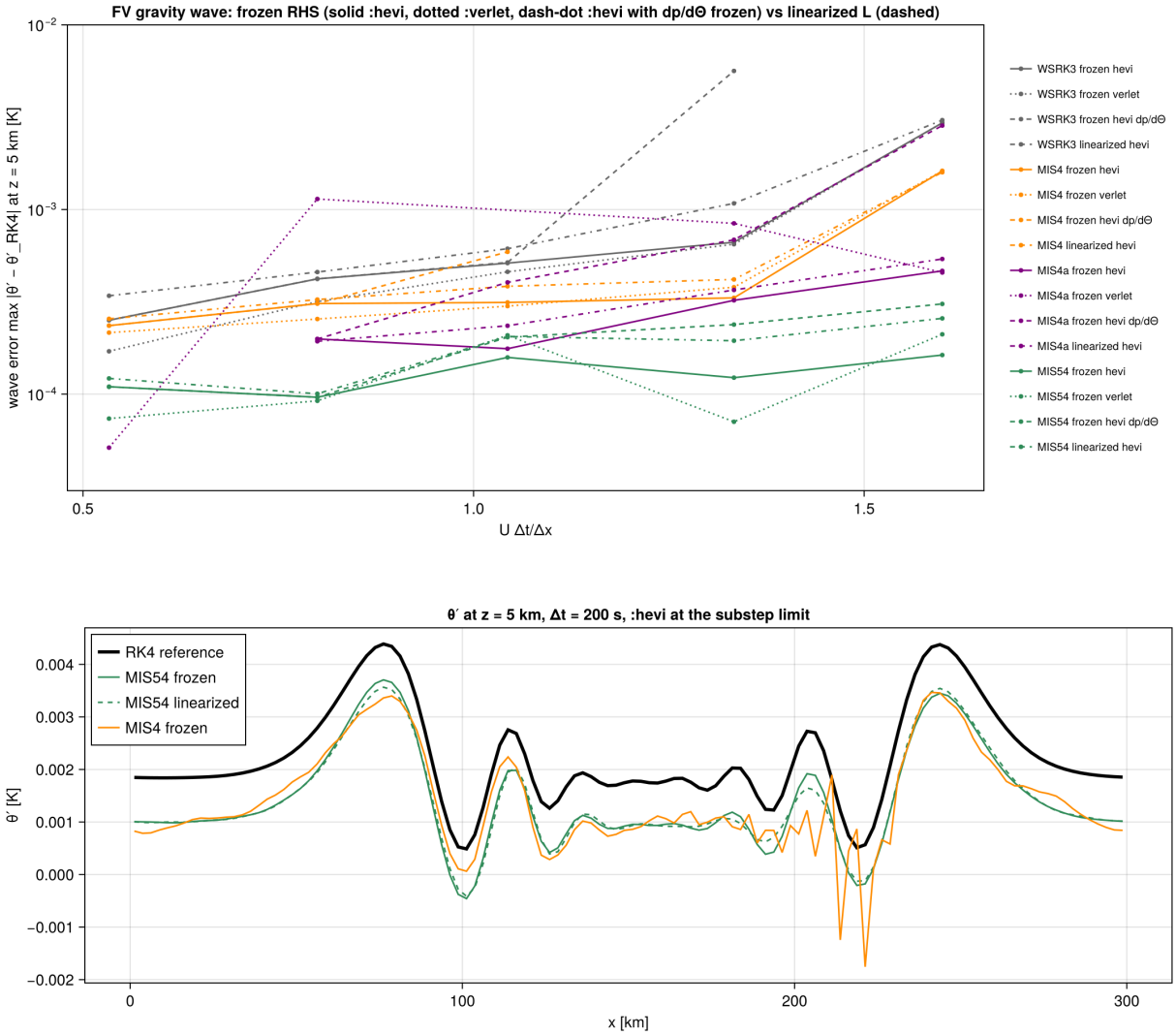
	additive $F = S(y) + G(z)$	frozen coefficient
WSRK3, MIS2, MIS3C	2	2
MIS4, MIS4a	3	**2**
MIS54	4	**2**
MIS_KW3 (Knoth–Wolke)	3	3

With a non-additive F the mixed trees ($F^{(2)}_{(1,1)}FF, \dots$) add order conditions that MIS4/4a/54 do not satisfy; methods with time-dependent weights a^p_{ij} satisfying them are to be constructed with the B-series program.

Gravity wave, 300 km (wave error in K, n_{sub} at the substep limit):

Δt [s] ($U\Delta t/\Delta x$)	67 (0.53)	100 (0.8)	130 (1.04)	167 (1.33)	200 (1.6)
MIS54 frozen `hevi`	1.1e-4	9.6e-5	1.6e-4	1.2e-4	**1.6e-4**
MIS54 frozen `verlet`	7.4e-5	9.2e-5	2.1e-4	7.1e-5	2.1e-4
MIS54 linearized `hevi`	1.1e-4	9.6e-5	2.0e-4	2.4e-4	3.1e-4
MIS4 frozen `hevi`	2.3e-4	3.1e-4	3.1e-4	3.3e-4	1.6e-3
MIS4 linearized `hevi`	2.3e-4	3.1e-4	5.9e-4	2.7e-2	unst.
WSRK3 frozen `hevi`	2.5e-4	4.2e-4	5.1e-4	6.6e-4	2.9e-3
WSRK3 linearized `hevi`	2.5e-4	4.2e-4	5.2e-4	5.6e-3	unst.

At $U\Delta t/\Delta x \lesssim 1$ both splits agree; at larger steps the frozen split stays stable and accurate where the linearized one degrades or blows up, and MIS54 halves its error at 200 s. A plausible reason: the linearized split puts the transport of Θ and of the pressure perturbation by the slow velocity into S and couples it explicitly to the acoustics; the frozen split keeps all mass and Θ fluxes in the fast loop with the full U. MIS4a fails at single step sizes in both splits; MIS_KW3 is stable only at small steps.



13 Open questions and next steps

- Frozen split (Section 12): MIS coefficients of order 3/4 for non-additive F, the frozen split for DGSEM and cut cells.

- a1 terrain FV: identify the terrain term in S (Section 8).
- MIS3C: second order with $L \neq 0$; compare with Knoth & Wensch (2014).
- Cut cells: warm layer in the lowest cells downstream of the hill.
- Gravity wave with a balanced initial state (no mean oscillation), DGSEM and the long channel; ETD methods with fewer fast integrations (shared $e^{TL}q^n$ of stages with equal c_i).
- A linear IMEX fast solver as in the MATLAB reference (one sparse LU of $I-\gamma\Delta\tau L_I$), wider choices of the implicit part (adaptive masks), retained HDG traces in 2D.

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